Experiment no : 02

Experiment : Determination of the extinction co-efficient of Cupra – ammonium complex , $(\mathbf{Cu(NH_3)})^{2+}$ and determination of the concentration of given unknown solution.

Theory:

According to Beer-Lambert's law.

$$\mathbf{A = \varepsilon Cl}$$

Where;

A=Absorbance (Optical density)

C= concentration of the solution

L= Light path length

ε=Extinction coefficient

Procedure:

1) Firstly, prepare a 0.1 $\mathrm{moldm^{-3}}$ $CuSO_4$ solution in a 100 mL volumetric flask.Transfer about 10.00 mL of 0.1 $\mathrm{moldm^{-3}}$ $CuSO_4$ solution into 50 mL volumetric flask and add 10.00 mL of conc NH_3. (Resulting solution is called as Cupra-ammonium stock solution).
2) Take 10.00 mL,8.00 mL, 6.00 mL, 5.00 mL, 4.00 mL and 2.00 mL of the above stock solution into 50.00 mL volumetric flasks using a burette and dilute them to 50.00 mL with distilled water.
3) Determine the best filter for the Cuppra-ammonium complex by measuring the absorbances of lowest and highest concentrations at each wave length.
4) Measure the absorbances of the above standard series solutions at the appropriate best filter.
5) Plot absorbance vs concentration ($\mathrm{moldm^{-3}}$)of Cu – ammonium complex .
6) Determine the extinction coefficient of the Cuppra –ammonium complex. Find the concentration of an unknown Cu^{2+} solution.(Take only 10.00 mL of the unknown Cu-ammonium complex solution)